

Remote Software Deployment and Malware Cleanup w/ Action1 - Help Desk Simulation

This walkthrough demonstrates how to simulate and resolve a common help desk security issue: a user reporting a suspicious file and potential malware on their workstation.

The goal of this exercise is to practice:

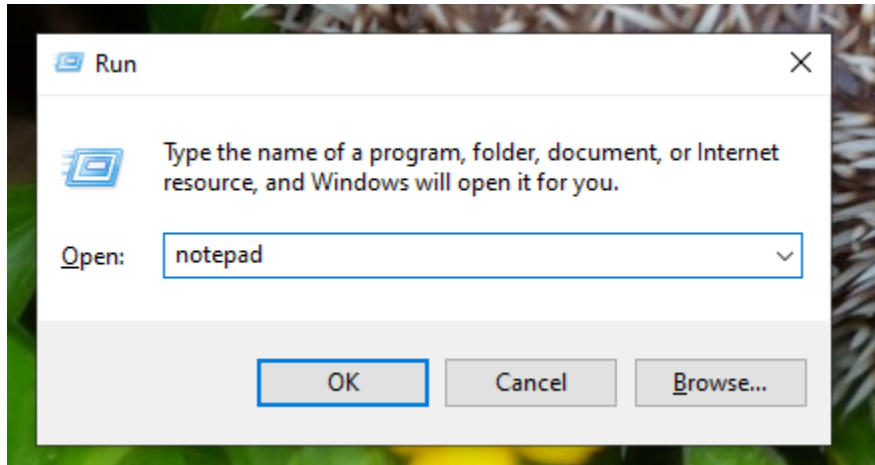
- Safely simulating a malware-related incident on a Windows 10 workstation using an EICAR test file
- Collecting user-reported symptoms through a help desk ticketing system (ServiceNow)
- Verifying endpoint status and inventory using an RMM/endpoint management tool (Action1)
- Remotely deploying security software to a single endpoint
- Running a malware scan and performing threat cleanup
- Verifying endpoint health after remediation
- Writing clear, professional help desk ticket documentation

This lab reflects realistic Tier 1–Tier 2 help desk responsibilities involving endpoint management, incident response, and user communication in a controlled environment.

Step 1 – Simulate Malware on the Employee Workstation VM

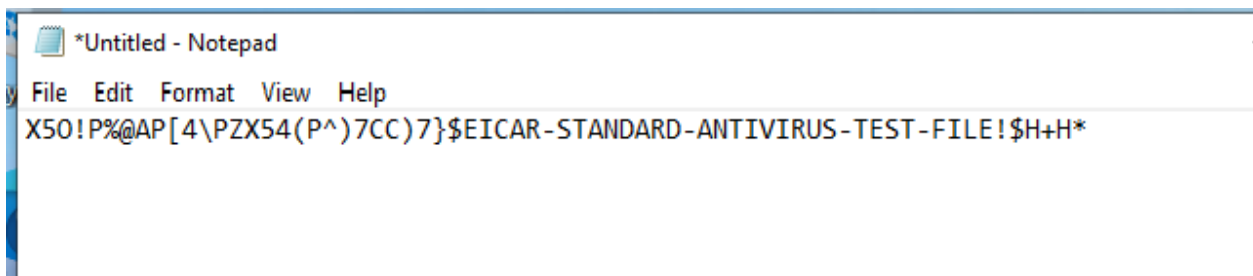
On your **Windows 10 VM**:

1. Press Windows + R → Type notepad → Enter



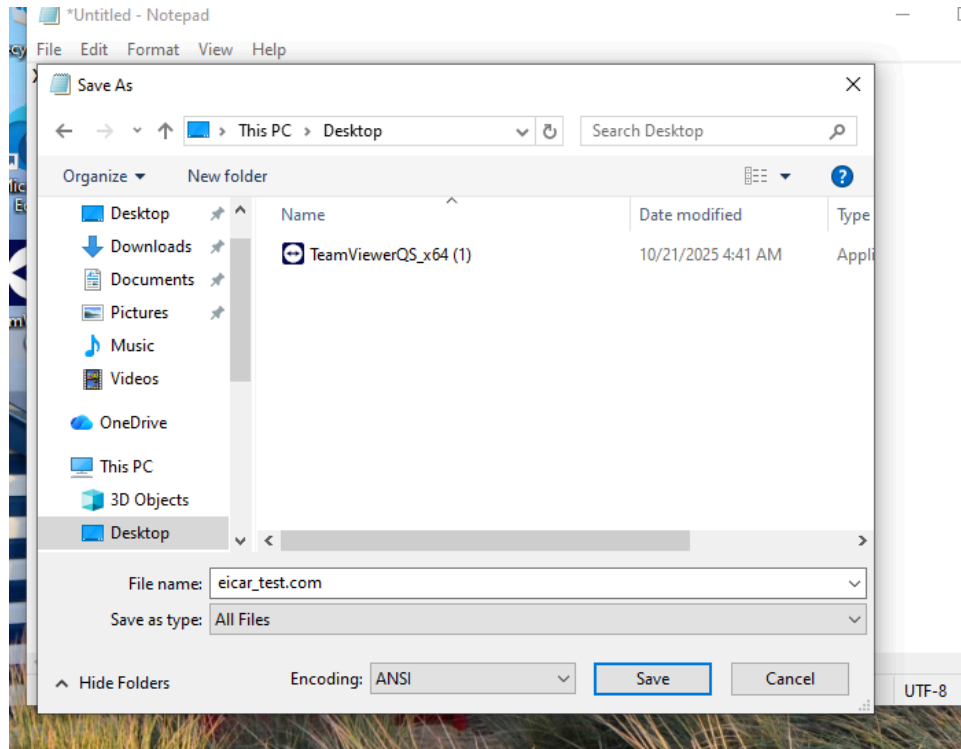
2. Use the EICAR Test String (used to test antivirus tools and not actual malware, but antivirus tools treat it as malware). Paste this exact line into the notepad file:

`X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*`



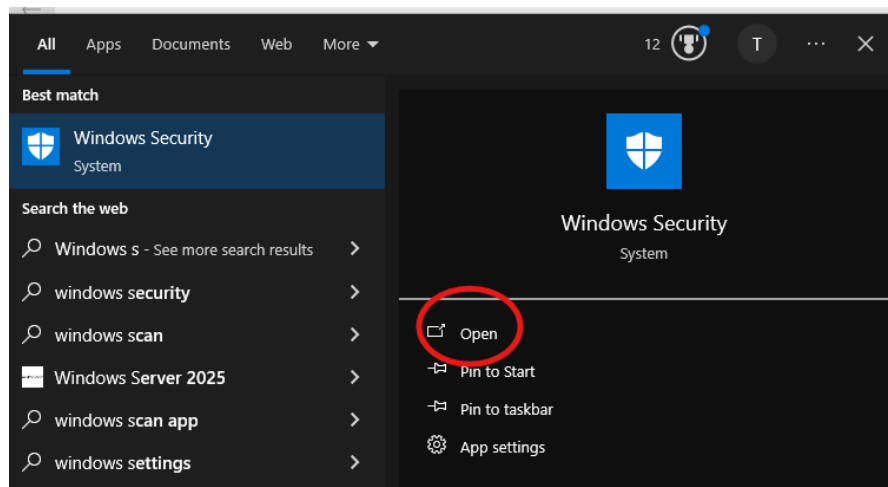
3. Save the file by clicking **File** → **Save As**.
 - Location: **Desktop**
 - File name: **eicar_test.com**

- Save as type: **All Files**
- Encoding: **ANSI**

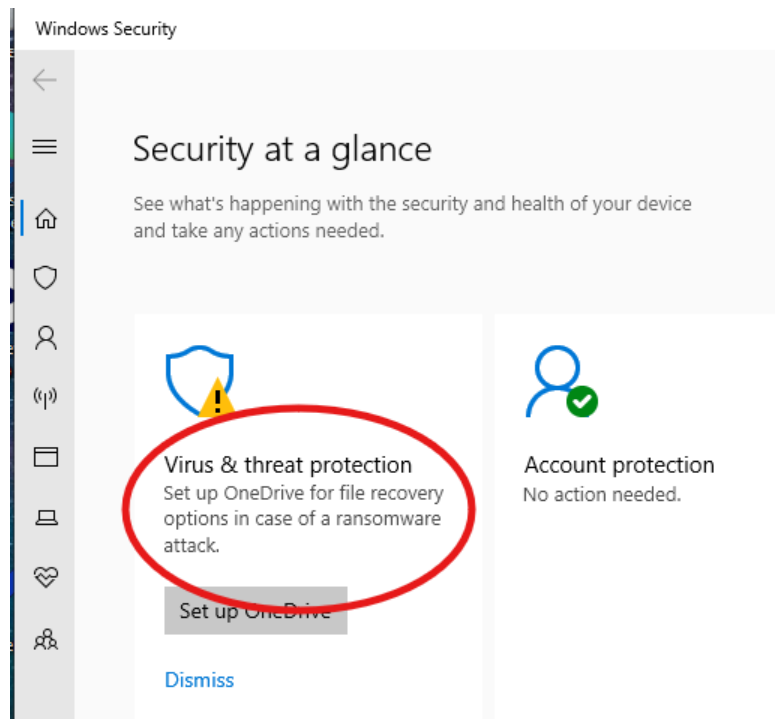


4. Click **Save**.

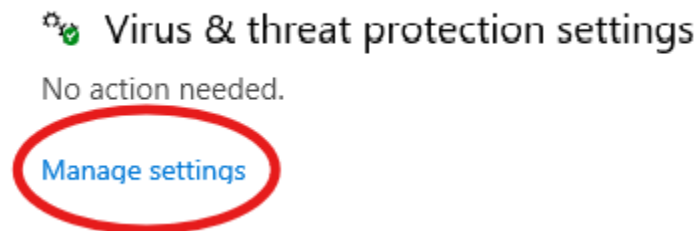
5. Sometimes **Windows Defender** will automatically remove the EICAR test file, so you should add an exclusion. Go to **Windows Security**.



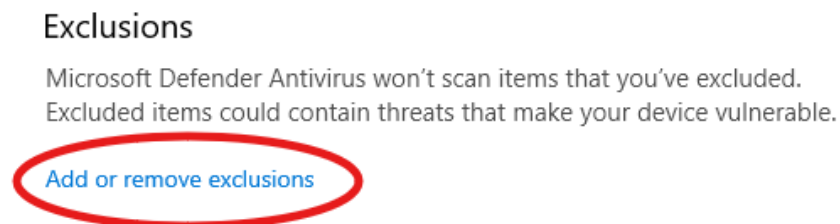
6. Go to **Virus & threat protection**.



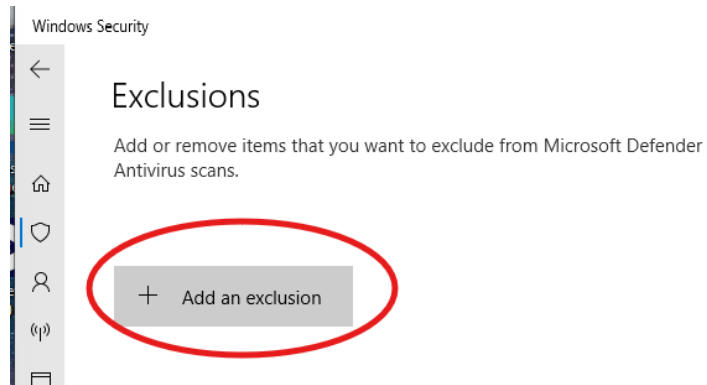
7. Click **Manage settings**.



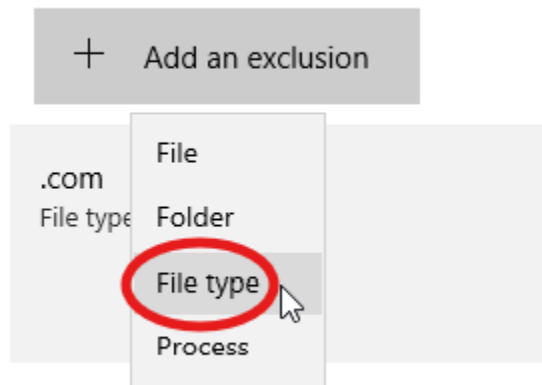
8. Scroll to **Exclusions** and click **Add or remove exclusions**.



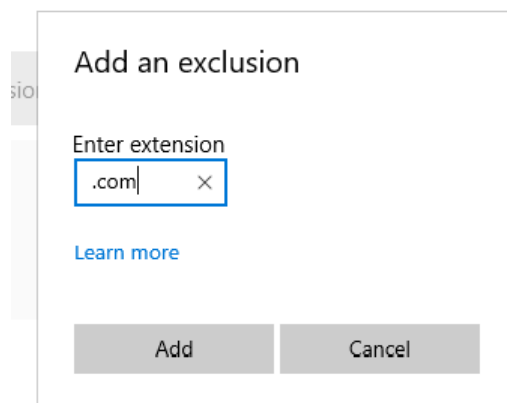
9. Click **Add an exclusion**.



10. Click **File type**.



11. Type the extension of your EICAR test file and click **Add**.

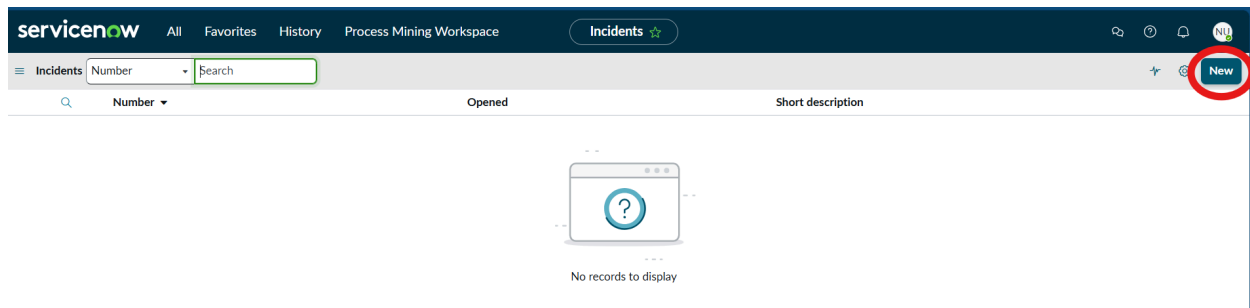


Now, Windows Defender won't remove the EICAR file and Malwarebytes will detect this "malware".

Step 2 – Create the help-desk ticket as the user and confirm the ticket as the systems administrator

On **ServiceNow** logged in as the employee user:

1. Open your ticketing system ([ServiceNow](#)) and create a new ticket to report the issue:



- Urgency: **2-Medium**
- More information: *There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware. I didn't download it. I don't know how it got there. Please check it out.*

There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware. I didn't download it. I don't know how it got there. Please check it out.

There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware.

1. Go to **Incidents** and locate the new ticket from the employee.

servicenow All Incidents Vie... ☆ 🔍 🌐 💬 📄 ⋮ 👤			
☰ 🔍 Incidents View: Self Service Number Search ⚡ ⚙️ Actions on selected rows... New			
All			
☐	🔍 Number ▾	Opened	Short description
☐	INC0010002	2025-12-07 08:21:39	There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware.
	INC0010001	2025-11-25 00:24:42	Locked out of domain account. Please help!

2. Change State to In Progress and then click Update

servicenow

All

Incident - INC... ☆

🔍

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👤

<

☰

Incident

INC0010002

View: Self Service

🔗

⚡

⚙️

⋮

Discuss

↑

↓

Follow

Update

Resolve

Delete

Number

INC0010002

Opened

2025-12-07 08:21:39

📅

* Caller

Naruto Uzamaki

🔍

🗣️

📄

Closed

📅

Watch list

🔒

👤

Urgency

2 - Medium

▾

State

In Progress

▾

* Short description

There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on

💡

Related Search Results

>

Activities: 3

👤 System Administrator

Comments • 2025-12-09 06:13:53

I will investigate this matter and get back to you as soon as possible. Thank you Naruto!

NU Naruto Uzamaki

Comments • 2025-12-07 08:21:39

There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware. I didn't download it. I don't know how it got there. Please check it out.

NU Naruto Uzamaki

Field changes • 2025-12-07 08:21:39

Impact

3 - Low

Incident state

New

Opened by

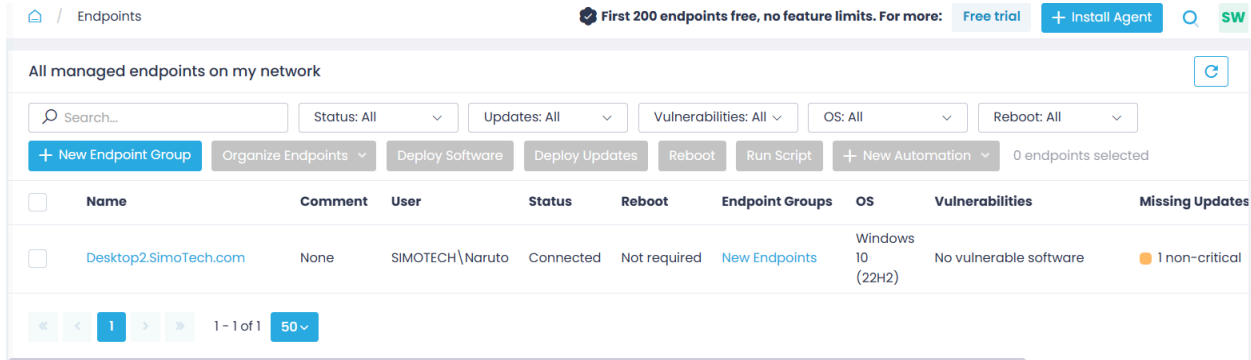
Naruto Uzamaki

Priority

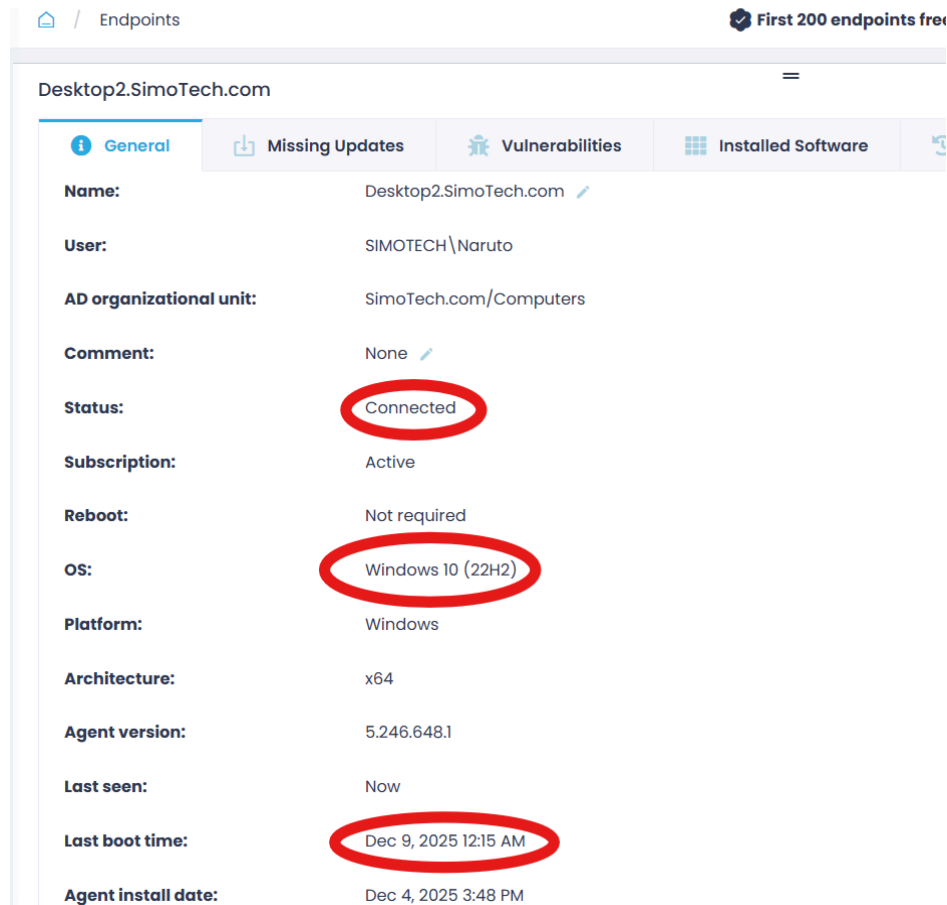
4 - Low

Step 3 – Confirm endpoint state in Action1

1. In Action1 → Endpoints, find Desktop2.SimoTech.com



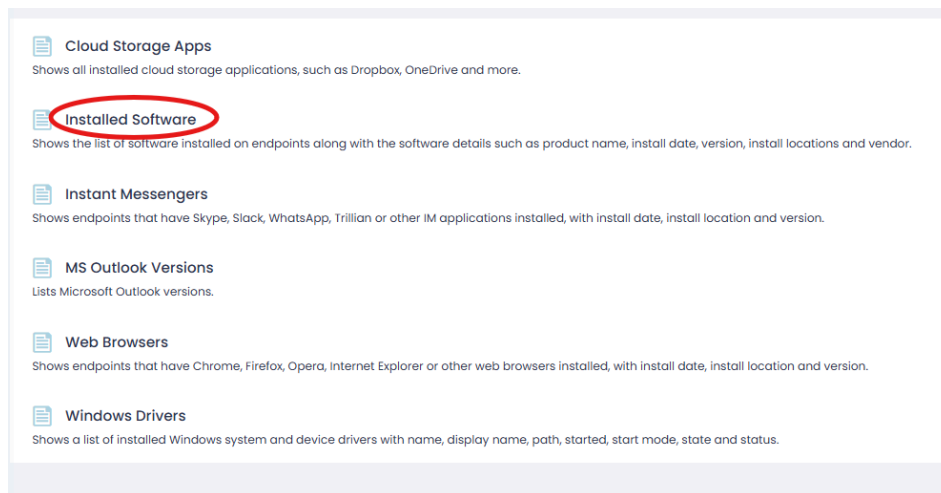
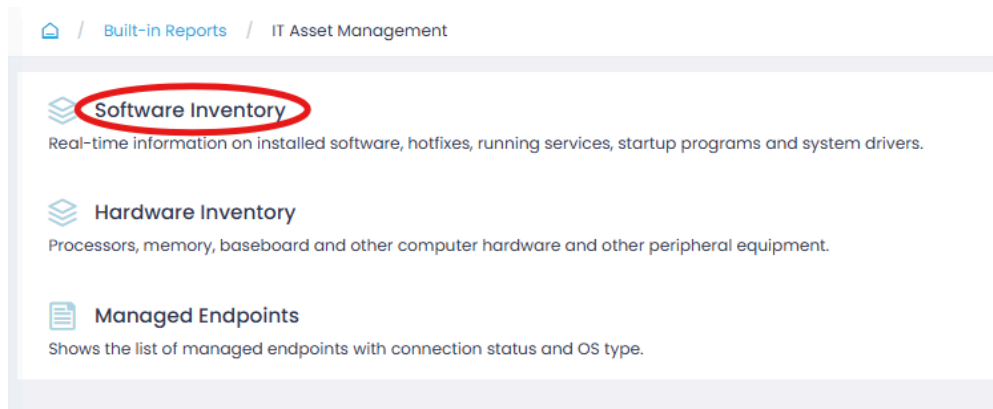
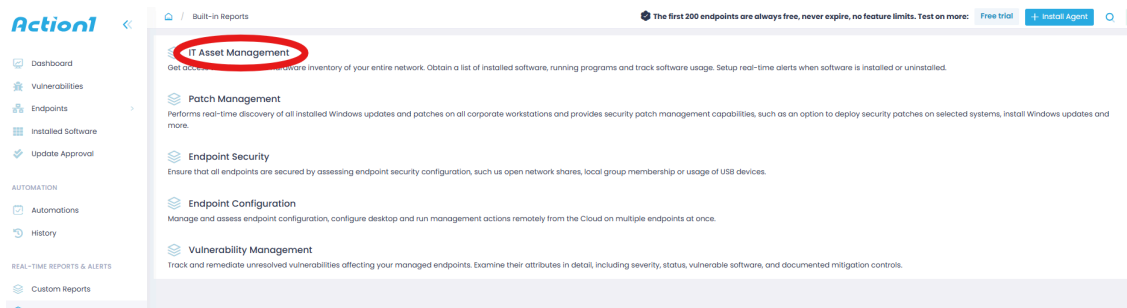
2. Verify status **Connected**, **OS**, **IP**, and **Last boot time**.



IP address:

10.0.2.4

3. Pull quick inventory: Installed software, running processes, disk free space.



[Home](#) / [Built-in Reports](#) / [IT Asset Management](#) / [Software Inventory](#) / [Installed Software](#)
First 200 endpoints free, no

Shows the list of software installed on endpoints along with the software details such as product name, install date, version, install locations and vendor.

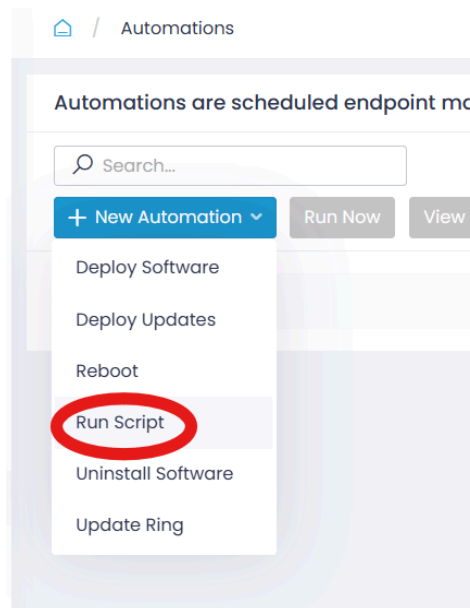
Responses: Live and Cached ▾
Expand details: No ▾

Name	Details
Action! Agent	1
Microsoft Edge	1
Microsoft OneDrive	3
Microsoft Update Health Tools	1
Update for x64-based Windows Systems (KB5001716)	1

« < 1 > »
1 - 5 of 5
50 ▾

4. Run this short Process check via **Run Script** (PowerShell):

*Get-Process | Sort-Object CPU -Descending | Select-Object -First 10 |
 Format-Table -AutoSize Get-EventLog -LogName Application -Newest 20 |
 Select TimeGenerated, EntryType, Source, Message*



Script to execute. Insert an ad-hoc script or select a script from [Script Library](#):

Get-Process | Sort-Object CPU -Descending | Select-Object -First 10 | Format-Table -AutoSize
Get-EventLog -LogName Application -Newest 20 | Select TimeGenerated, EntryType, Source, Message

Type your script manually or [upload](#) a script file [?](#)

The script shall exit with exit code 0 on success or non-zero exit code if it encounters any errors.

Initiate a reboot if the script returns any of these exit codes:

Script language:

PowerShell - Windows

Save in Script Library

If you plan to reuse this script in your other actions, it's best to save it in Script Library and reference later

The automation will be run on one or more specified endpoints:

Add Endpoints

Remove All

Name	Type	Status	Actions
Desktop2.SimoTech.com	Endpoint	Connected	

<<

<

1

>

>>

1 - 1 of 1

50

Previous

Cancel

Next Step

Automation name:

Process Check

Select when do you want to run this automation:

☒ Run once

☒ Run now

☐ At specified time:

12/09/2025

01:24 AM

☐ Every

☐ Weekly

☐ Monthly

☐ No schedule yet

Missed schedule retry and maintenance window

Automation completion deadline:

24

hours

after the scheduled start time

Extending this parameter ensures that offline or disconnected endpoints will retry the automation once they come online within the specified timeframe. The automation will be terminated if it exceeds this deadline, ensuring timely completion.

[See examples](#)

Previous

Cancel

Finish

5. → Review output for suspicious processes or errors.

Automation History / Automation Details

The first 200 endpoints are always free, never expire, no feature limits. Test on more: [Free trial](#)

Automation "Process Check" details

Search...

Status: All

Endpoint	Operation	Date/Time	Status	Details
				Handles NPM(k) PM(k) WS(k) CPU(s) Id SI ProcessName ----- 938 226 311752 262076 94.86 2944 0 MsMpEng 2791 0 196 148 68.05 4 0 System 868 85 73368 74448 59.75 4904 0 svchost 238 15 58528 61964 12.92 1960 0 svchost 1149 18 7162 14748 10.30 812 0 svchost 615 29 25400 40892 6.22 2764 0 svchost 388 21 7096 23532 5.94 2740 0 actionl_agent 664 11 4976 10056 4.78 588 0 services 212 87 58540 142328 4.52 4604 1 explorer 492 28 23096 33420 4.30 3608 0 svchost TimeGenerated EntryType Source Message ----- 12/9/2025 10:05:52 AM Information SecurityCenter Updated Windows Defender status successfully ... 12/9/2025 10:05:52 AM Information SecurityCenter Updated Windows Defender status successfully ... 12/9/2025 10:05:52 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 10:05:52 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 10:15:58 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 10:15:58 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 10:15:58 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 12:56:27 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 12:56:50 AM Information SecurityCenter Updated Windows Defender status successfully ... 12/9/2025 12:56:29 AM Error Software Protection Platform Service License Activation (slu.exe) failed with the... 12/9/2025 12:56:29 AM Information Software Protection Platform Service The Software Protection service has completed... 12/9/2025 12:56:20 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 12:47:20 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 12:46:21 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 12:37:17 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 12:36:31 AM Information Software Protection Platform Service Offline downlevel migration succeeded. 12/9/2025 12:29:12 AM Information SecurityCenter Updated Windows Defender status successfully ... 12/9/2025 12:27:48 AM Information Software Protection Platform Service Successfully scheduled Software Protection se... 12/9/2025 12:26:26 AM Information SecurityCenter Updated Windows Defender status successfully ... 12/9/2025 12:26:24 AM Information ESENT svchost (3684D,50) DS_Token_DB: The database... 12/9/2025 12:26:24 AM Information ESENT svchost (3684D,0) DS_Token_DB: The database ...
Desktop2.SimoTech.com	Completed	Dec 9, 2025 1:10 AM	Success	

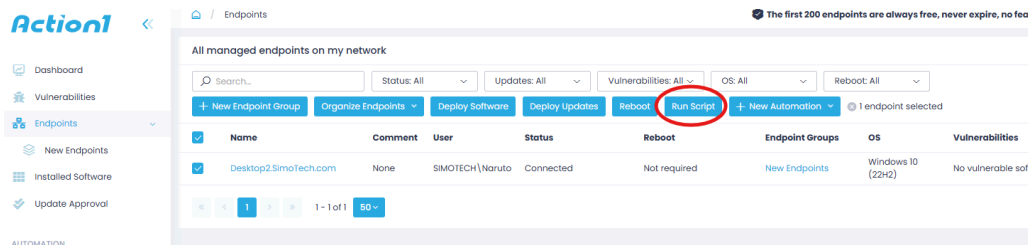
1 - 1 of 1 50

Based on the output, there are no suspicious processes related to potential malware.

Step 4 – Prepare the installer and install Malwarebytes Remotely to the Endpoint using Action1

Use vendor download URL. Use the official Malwarebytes download link (https://downloads.malwarebytes.com/file/mb4_offline) and let the endpoint download directly. This avoids hosting files and is simpler than other alternatives.

1. Go to **Endpoints**, click the box of the target machine, and then click **Run Script**.



2. Change the script language to **Command - Windows**.

1 Run Script 2 Select Endpoints 3 Schedule

Script to execute. Insert an ad-hoc script or select a script from [Script Library](#):

Type your script manually or [upload a script file](#)

The script shall exit with exit code 0 on success or non-zero exit code if it encounters any errors.
Initiate a reboot if the script returns any of these exit codes:

Script language:
Command - Windows

3. Copy and paste the script below into the form, **Script to execute**.

Script to execute. Insert an ad-hoc script or select a script from [Script Library](#):

```
powershell -Command "New-Item -ItemType Directory -Path C:\Temp -Force"
powershell -Command "Invoke-WebRequest -Uri https://downloads.malwarebytes.com/file/mb4_offline -OutFile |
C:\Temp\MalwarebytesInstaller.exe
C:\Temp\MalwarebytesInstaller.exe /silent
```

Type your script manually or [upload](#) a script file [?](#)

powershell -Command "New-Item -ItemType Directory -Path C:\Temp -Force"

- The first command opens **PowerShell** and creates a new directory called **C:\Temp**. We create the directory because we need a place to store the installer file. “**-Force**” makes it so it has to work: either create the folder if it doesn’t exist or not fail if it already exists.

powershell -Command "Invoke-WebRequest -Uri https://downloads.malwarebytes.com/file/mb4_offline -OutFile C:\Temp\MalwarebytesInstaller.exe"

- The second command uses **PowerShell** to download the installer file directly from the vendor site and then saves it as **MalwarebytesInstaller.exe**. This allows the endpoint to install the software by itself without the use of file sharing, USB, or manual download.

C:\Temp\MalwarebytesInstaller.exe /silent

- The last command runs the Malwarebytes installer. **/silent** makes the installation happen in the background, with no popups, so the employee can continue their work on their machine without interruption.

4. After inputting the script then click **Next Step**.

Script to execute. Insert an ad-hoc script or select a script from [Script Library](#):

```
powershell -Command "New-Item -ItemType Directory -Path C:\Temp -Force"
powershell -Command "Invoke-WebRequest -Uri https://downloads.malwarebytes.com/file/mb4_offline -OutFile
C:\Temp\Malwarebytesinstaller.exe"
C:\Temp\Malwarebytesinstaller.exe /silent
```

Type your script manually or [upload](#) a script file [?](#)

The script shall exit with exit code 0 on success or non-zero exit code if it encounters any errors.

Initiate a reboot if the script returns any of these exit codes:

Script language:

Command - Windows

[Save in Script Library](#) If you plan to reuse this script in your other actions, it's best to save it in Script Library and reference later

☐ Execute the script above only if the automation condition script returns a non-zero error code

[Reboot options](#): automatically, show the message, message timeout 4 hours

[Cancel](#) [Next Step](#)

5. Make sure the target endpoint is selected and click **Next Step**.

1 Run Script 2 **Select Endpoints** 3 Schedule

The automation will be run on one or more specified endpoints:

[Add Endpoints](#) [Remove All](#)

Name	Type	Status	Actions
Desktop2.SimoTech.com	Endpoint	Connected	

« < 1 > » 1 - 1 of 1 50 ▾

[Previous](#) [Cancel](#) [Next Step](#)

6. Name the automation and make sure it's set to **Run now**. Click **Finish**.

1 Run Script

2 Select Endpoints

3 Schedule

Automation name:

Malwarebytes Background Installation

Select when do you want to run this automation:

☒ Run once

☒ Run now

☐ At specified time:

12/09/2025

10:01 PM

Missed schedule retry and maintenance window

Automation completion deadline:

24

hours

 after the scheduled start time ?

Extending this parameter ensures that offline or disconnected endpoints will retry the automation once they come online within the specified timeframe. The automation will be terminated if it exceeds this deadline, ensuring timely completion.

[See examples](#)

Previous

Cancel

Finish

7. Confirm that the script is **Running** and wait for it to finish. After some minutes, the status will be **Completed**.

Automation History

Automation Details

The first 200 endpoints are al

Automation "Malwarebytes Background Installation" details

Search...

Status: All

Endpoint	Operation	Date/Time	Status	Details
Desktop2.SimoTech.com	Start Automation	Dec 9, 2025 9:49 PM	Running	Waiting for the endpoint to run the automation.

1

1 - 1 of 1

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Automation History

The first 200 endpoints are always free, never expire, no feature limits. Test on more: [Free trial](#) [+ Install Agent](#) [Q](#) [SW](#)

Automation execution history

Search... Completion: All

+ New Automation Stop 0 automations selected

Name	Status	Started	Finished	Endpoints	Actions
Malwarebytes Background Installation	Completed	Dec 9, 2025 9:49 PM	Dec 9, 2025 9:53 PM	Desktop2.SimoTech.com	View Details
Process Check	Completed	Dec 9, 2025 1:10 AM	Dec 9, 2025 1:10 AM	Desktop2.SimoTech.com	
ipconfig	Completed	Dec 6, 2025 12:25 AM	Dec 6, 2025 12:25 AM	Desktop2.SimoTech.com	

1 - 3 of 3 50

8. Click **View Details** in the far right and confirm it is a **Success**.

Automation execution history

Search...

Completion: All

+ New Automation

Stop

0 automations selected

☐	Name	Status	Started	Finished	Endpoints	Actions
☐	Malwarebytes Background Installation	Completed	Dec 9, 2025 9:49 PM	Dec 9, 2025 9:53 PM	Desktop2.SimoTech.com	⋮ View Details
☐	Process Check	Completed	Dec 9, 2025 1:10 AM	Dec 9, 2025 1:10 AM	Desktop2.SimoTech.com	
☐	ipconfig	Completed	Dec 6, 2025 12:25 AM	Dec 6, 2025 12:25 AM	Desktop2.SimoTech.com	⋮

Automation History

Automation Details

The first 200 endpoints are always free, never expire, no feature limits. Test on more:

Free trial

+ Install Agent

SW

Automation "Malwarebytes Background Installation" details

Search...

Status: All

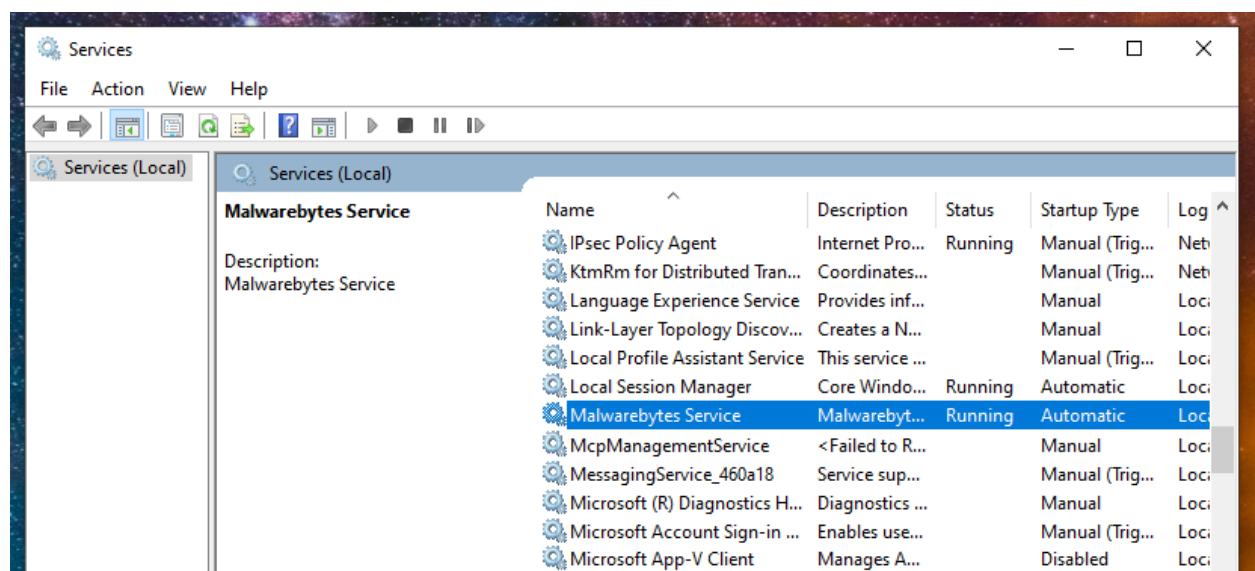
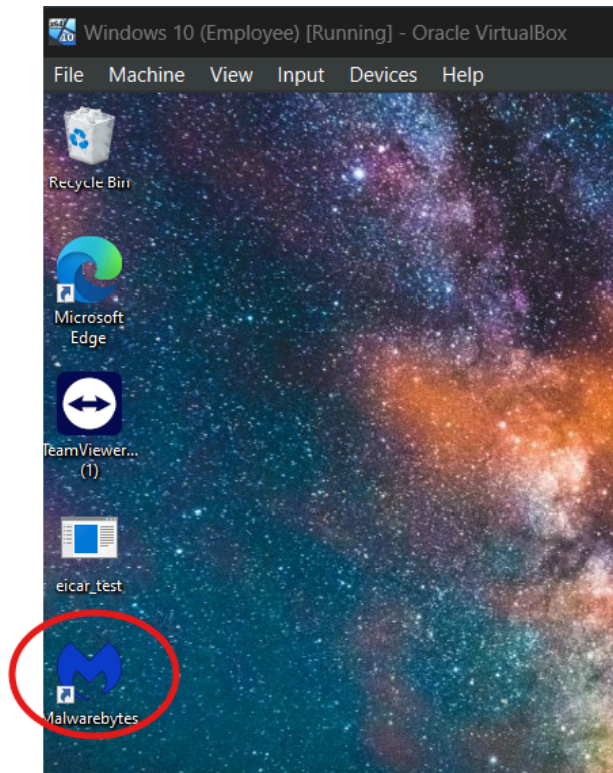
Endpoint	Operation	Date/Time	Status	Details								
▼ Desktop2.SimoTech.com	Completed	Dec 9, 2025 9:53 PM	Success	C:\Windows\system32>powershell -Command "New-Item -ItemType Directory -Path C:\Temp -Force" Directory: C:\ <table><thead><tr><th>Mode</th><th>LastWriteTime</th><th>Length</th><th>Name</th></tr></thead><tbody><tr><td>d----</td><td>12/9/2025 9:49 PM</td><td></td><td>Temp</td></tr></tbody></table> C:\Windows\system32>powershell -Command "Invoke-WebRequest -Uri https://downloads.malwarebytes.com/file/mb4_offline -OutFile C:\Temp\Malwarebytesinstaller.exe" C:\Windows\system32\C:\Temp\Malwarebytesinstaller.exe /silent	Mode	LastWriteTime	Length	Name	d----	12/9/2025 9:49 PM		Temp
Mode	LastWriteTime	Length	Name									
d----	12/9/2025 9:49 PM		Temp									

1

1 - 1 of 1

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9. Go to the Windows 10 VM and confirm that Malwarebytes was successfully installed. Check the **Desktop** and you should see the shortcut. Go to **Services** and you should see it running.

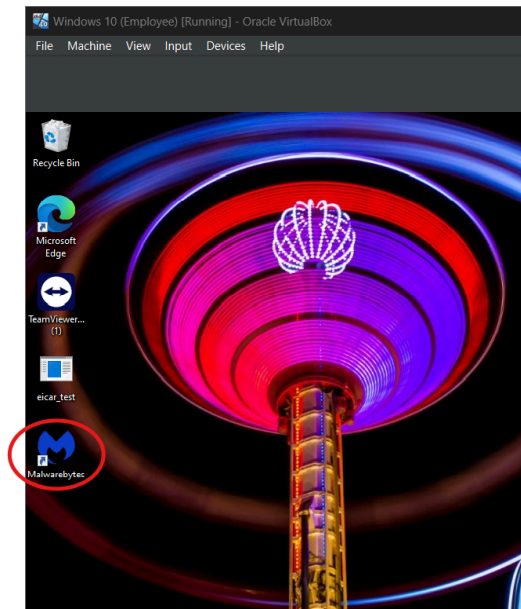


Malwarebytes has been successfully remotely installed by Action1 in the employee's machine.

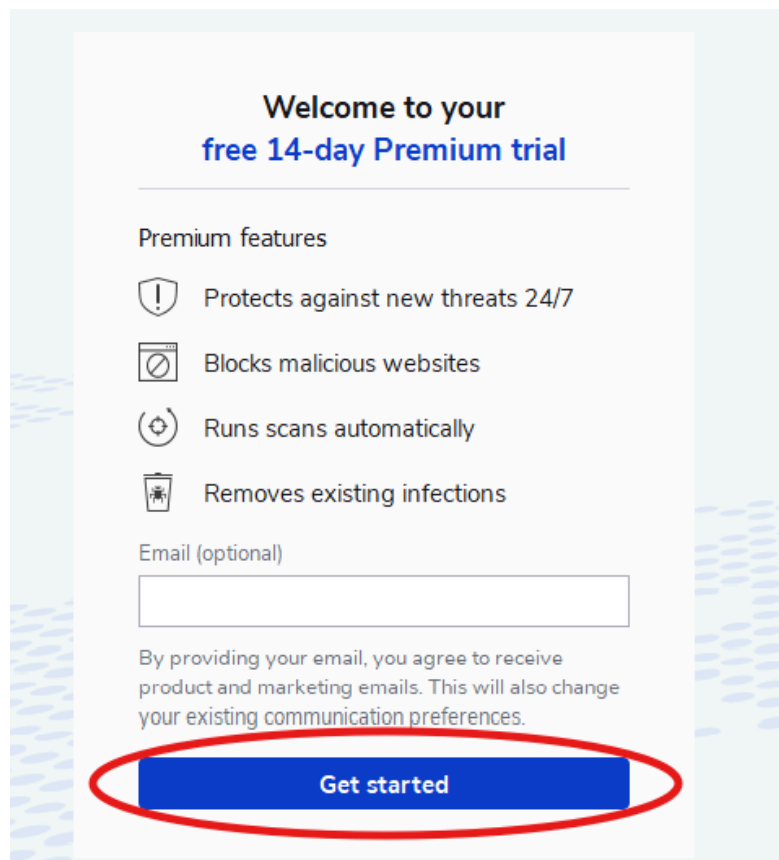
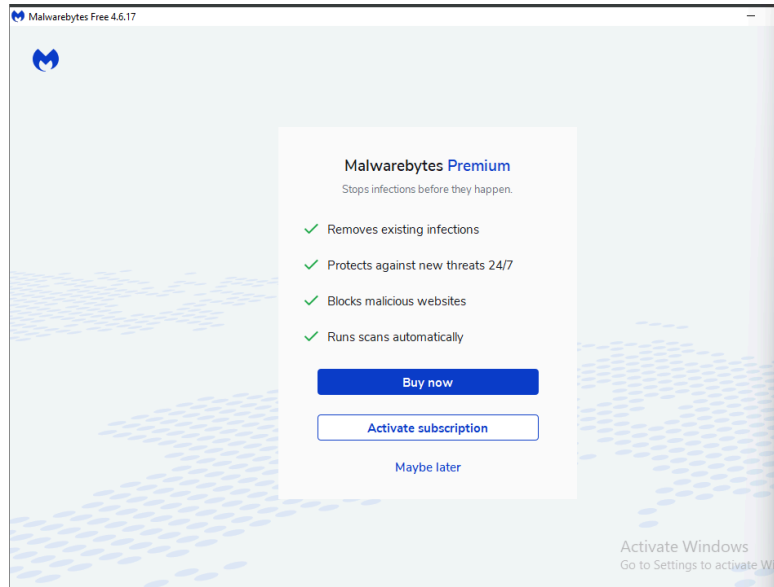
Step 5 — Run Malwarebytes Scan & Quarantine the Threat

On the **Windows 10 Employee VM (Desktop2/Simotech.com)**:

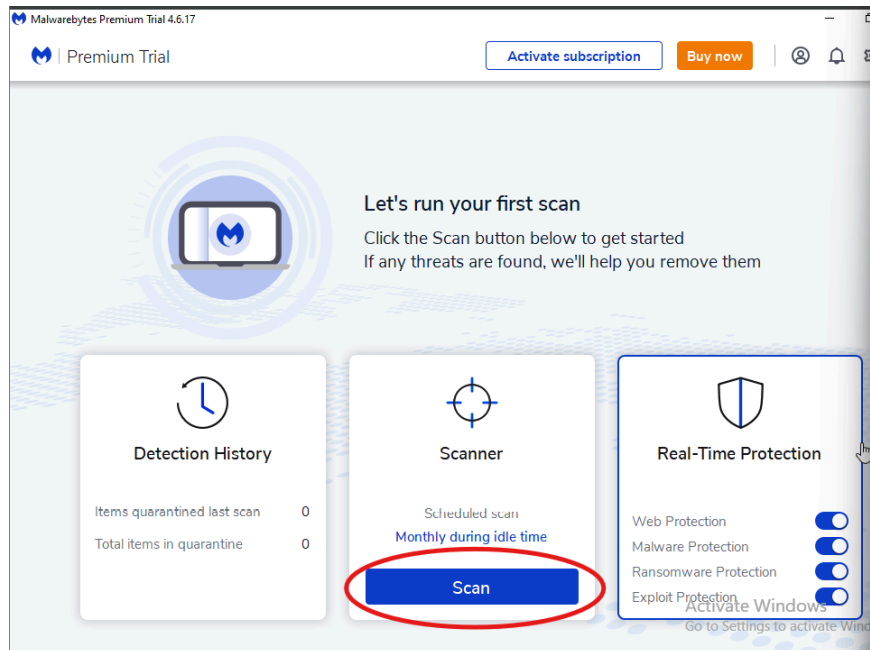
1. Open **Malwarebytes** from the Desktop shortcut.



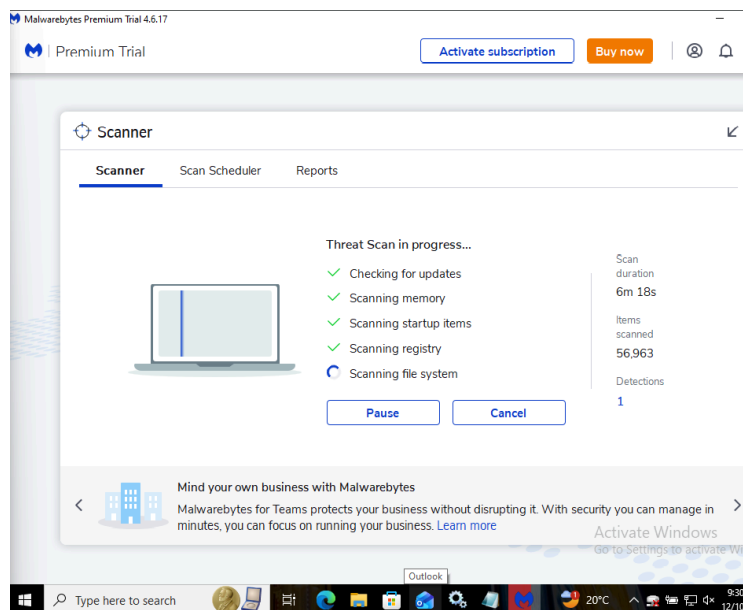
2. If prompted, skip sign-in or trial activation (free scan is sufficient). Click **Maybe later** and then **Get started**.



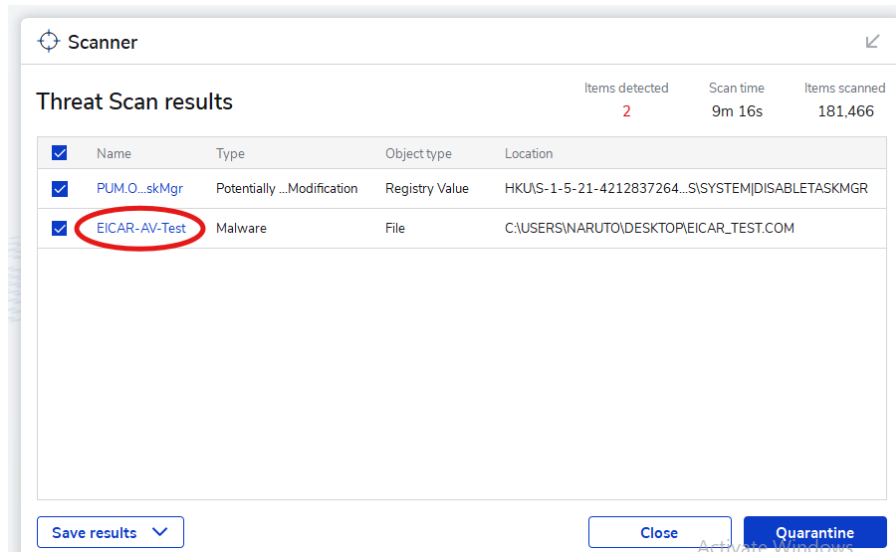
3. On the main dashboard, click **Scan**.



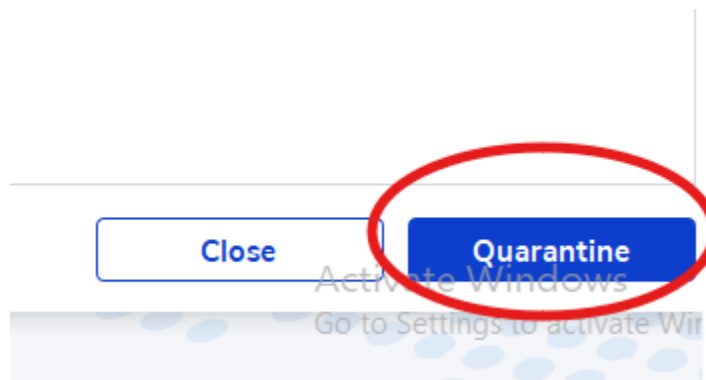
4. Allow Malwarebytes to run a **Threat Scan** (this may take a few minutes).



5. Malwarebytes should detect the **EICAR test file**.



6. Click **Quarantine** to safely remove the “malware”.



The simulated malware is now in quarantine.

Step 6 – Verify Cleanup & Endpoint Health

A. Verify the File Is Removed from the Desktop

On the Windows 10 VM:

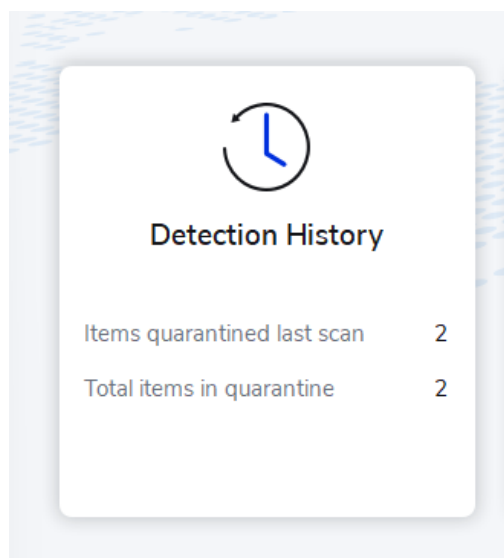
- Check the Desktop and confirm that **eicar_test.com** is gone.



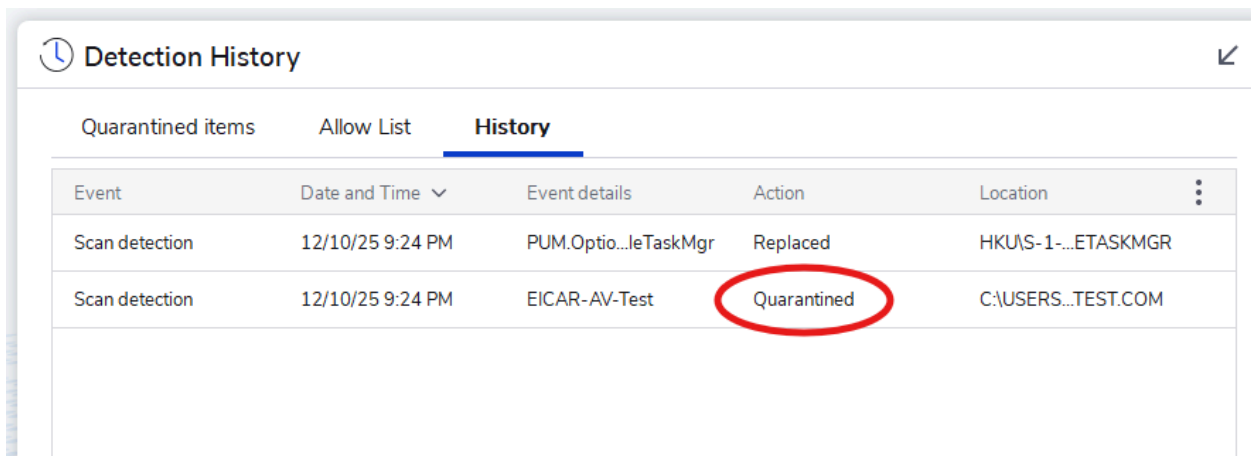
B. Verify in Malwarebytes

Inside Malwarebytes:

1. Go to **Detection History**



2. Confirm it is **Quarantined**.



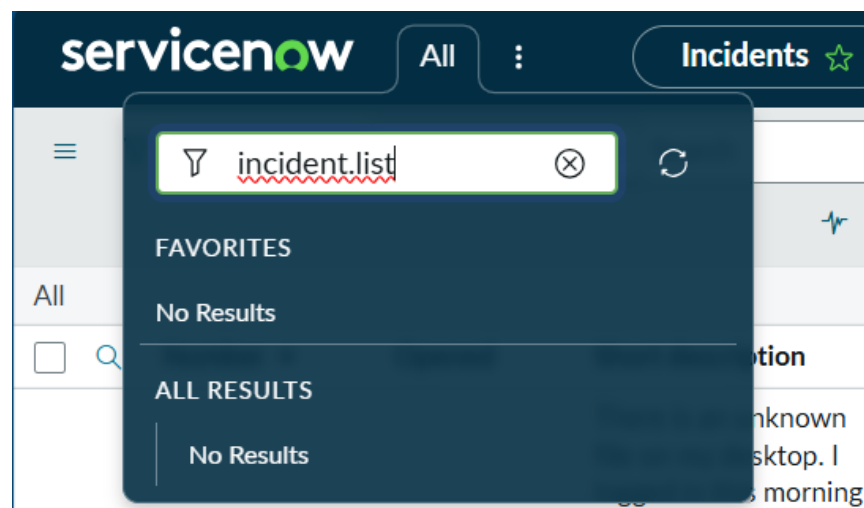
Detection History					
Quarantined items Allow List History					
Event	Date and Time	Event details	Action	Location	
Scan detection	12/10/25 9:24 PM	PUM.Optio...leTaskMgr	Replaced	HKU\S-1-...ETASKMGR	
Scan detection	12/10/25 9:24 PM	EICAR-AV-Test	Quarantined	C:\USERS...TEST.COM	

The endpoint is now verified clean.

Step 7 – Document Actions in ServiceNow & Resolve the Ticket

On **ServiceNow**, logged in as Systems Administrator:

1. Type **incident.list** and then open the relevant ticket.



All							
☐		Number ▼	Opened	Short description	Caller	Priority	State
☐		INC0010002	2025-12-07 08:21:39	There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware.	Naruto Uzamaki	4 - Low	New

2. Add the following notes in **Work Notes**:

- 1) *User reported an unknown file on desktop and possible malware.*
- 2) *Confirmed endpoint status via Action1.*
- 3) *Remotely deployed Malwarebytes using Action1 endpoint management.*
- 4) *Performed malware scan which detected a suspicious file.*
- 5) *Threat was successfully quarantined.*
- 6) *Verified endpoint is clean and operational.*

Activities: 4

System Administrator
 Work notes • 2025-12-10 06:49:44

User reported an unknown file on desktop and possible malware.
 Confirmed endpoint status via Action1.
 Remotely deployed Malwarebytes using Action1 endpoint management.
 Performed malware scan which detected a suspicious file.
 Threat was successfully quarantined.
 Verified endpoint is clean and operational.

System Administrator
 Comments • 2025-12-09 06:13:53

I will investigate this matter and get back to you as soon as possible. Thank you Naruto!

Naruto Uzamaki
 Comments • 2025-12-07 08:21:39

There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware. I didn't download it. I don't know how it got there. Please check it out.

Naruto Uzamaki
 Field changes • 2025-12-07 08:21:39

Impact	3 - Low
Incident state	New
Opened by	Naruto Uzamaki
Priority	4 - Low

3. Change **State** to **Resolved**

Channel

State

Impact

Urgency

[Priority](#)

Assignment group

Assigned to

on my desktop and am worried it might be malware.

on my desktop and am worried it might be malware. I didn't download

4. Resolution Code: **Resolved by Request** & Resolution Notes : **Security Issue – Malware Removed**

Notes Related Records Resolution Information

Knowledge ☒

* Resolution code

* Resolution notes

Resolved by

Resolved

5. Click **Update**

Incident - INC0010002

Resolved by

Resolved

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010002	2025-12-07 08:21:39	There is an unknown file on my desktop. I logged in this morning and saw a file called "eicar_test" on my desktop and am worried it might be malware.	Naruto Uzamaki	4 - Low	Resolved	Inquiry / Help	(empty)	(empty)	2025-12-10 06:55:13	admin

The ticket is now closed.

Final Outcome

- Malware was safely simulated using the EICAR test string
- Action1 was used to remotely deploy security software
- Malwarebytes successfully detected and removed the threat
- Endpoint health was verified
- Ticket was properly documented and resolved in ServiceNow